Project Title	DeployGuard: A Reliable Repository Scanning and Deployment Orchestration System
Supervisor Name	Yaseen Mushtaq
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Group Information	
Roll Number	**Student's Name**
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Module's Name	Module Description with Deliverables
1. User Authentication & Audit Logging	• Secure login, session handling, and role-based access are available for system users. • Login activity and important user actions are recorded for auditing.
2. Project & Workspace Management	• Users can create projects, connect a GitHub repository, and select the deployment directory. • Project details, deployment history, and current status are available in one workspace.
3. Rule-Based Stack Detection Engine	• Repository files are scanned to identify supported JavaScript and Python application structures. • The system detects the application runtime and port information needed for deployment.
4. DevOps Template Engine & Pre-Flight Validation	• Deployment settings are prepared from the detected project information. • Required files, commands, directory, and port settings are checked before deployment starts.
5. CI/CD Pipeline & Asynchronous Queuing	• Deployment operations are created, started, and tracked from beginning to completion. • GitHub Actions performs the build and deployment work while DeployGuard shows live status.
6. Security Scanning and Policy Enforcement	• Application dependencies and the container image are scanned for security issues. • Critical fixable findings are blocked while scan results are saved with the deployment.
7. Predictive Cost Analysis (FinOps)	• Estimated AWS infrastructure cost is generated from the Terraform deployment plan. • Cost information is linked with the project so users can review expected spending.
8. Infrastructure Provisioning & Service Discovery	• Terraform provisions the AWS resources required to run the application. • The deployed service endpoint and runtime information are saved after successful provisioning.
9. Distributed State Management	• Deployment and infrastructure state is stored so operations can be tracked reliably. • Terraform state is kept remotely in Amazon S3 with locking support.

Module's Name	Module Description with Deliverables
10. Persistent Storage & Backup Recovery	• Applications that need persistent files can use Amazon EFS storage. • Persistent application data remains separate from temporary container storage.
11. Zero-Downtime Container Orchestration & Event-Driven Rollbacks	• ECS Fargate runs application containers and manages service updates with health checks. • A deployment can be rolled back to a previous stable release when needed.
12. Dual-Stack Monitoring & Observability	• Application and deployment logs and metrics are available through the monitoring setup. • DeployGuard shows live deployment status and operational information for each project.
13. AI-Powered Log Analysis & Chatbot	• Deployment logs and failure information can be summarized into easier troubleshooting guidance. • The chatbot helps users understand deployment errors and project status.
14. Alerting & Notification	• Users receive alerts for important deployment events, failures, and status changes. • Amazon SNS is used for notification delivery where configured.
15. Infrastructure Lifecycle & Demolition (Destroy)	• Users can safely destroy deployed infrastructure through a confirmation-based process. • Cleanup is verified and the destroy result is recorded in DeployGuard.
16. Vendor Independence & Terraform Export	• Users can export the Terraform files used for their infrastructure. • The exported package gives users a portable copy of the infrastructure configuration.
17. Subscription & Billing	• DeployGuard supports Free and Pro subscription levels. • Stripe test-mode billing and plan-based feature access are handled by the platform.

Evaluator's Remarks / Suggestions

Respected Evaluator, if you recommend any improvement or suggest a new module, please mention it in this section.

Evaluator's Name	Remarks / Suggestions

Supervisor Signature: _____ Signature FYP Committee Member: _____